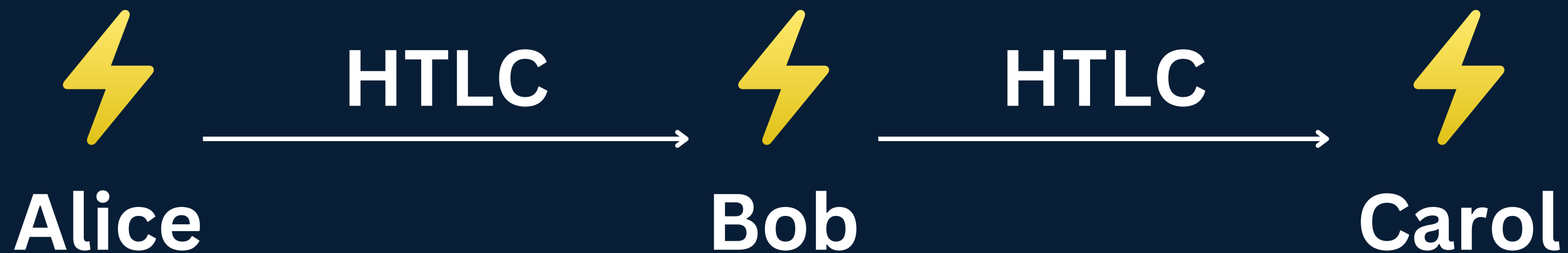




Lightning

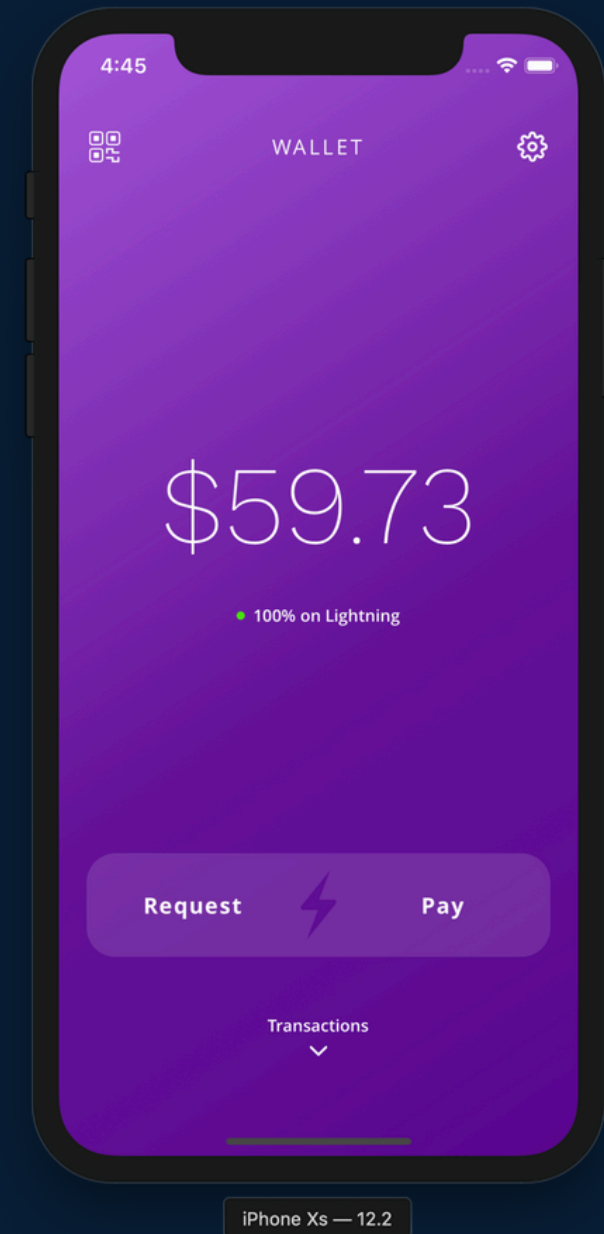
- **First Layer 2 on Bitcoin**
- **Payment channel network**

Payments on Lightning



Lightning on mobile

- Chain sync
- Public channel gossip
- Backups
- Liquidity



Chain sync

- Full node not feasible
- Light client? (BIP-157)
- Electrum protocol

Public channel gossip

- Torrent-like
- Rapid gossip sync?
- Trampoline routing

Backups

- Old channel state → loss of funds
- Cloud backups?

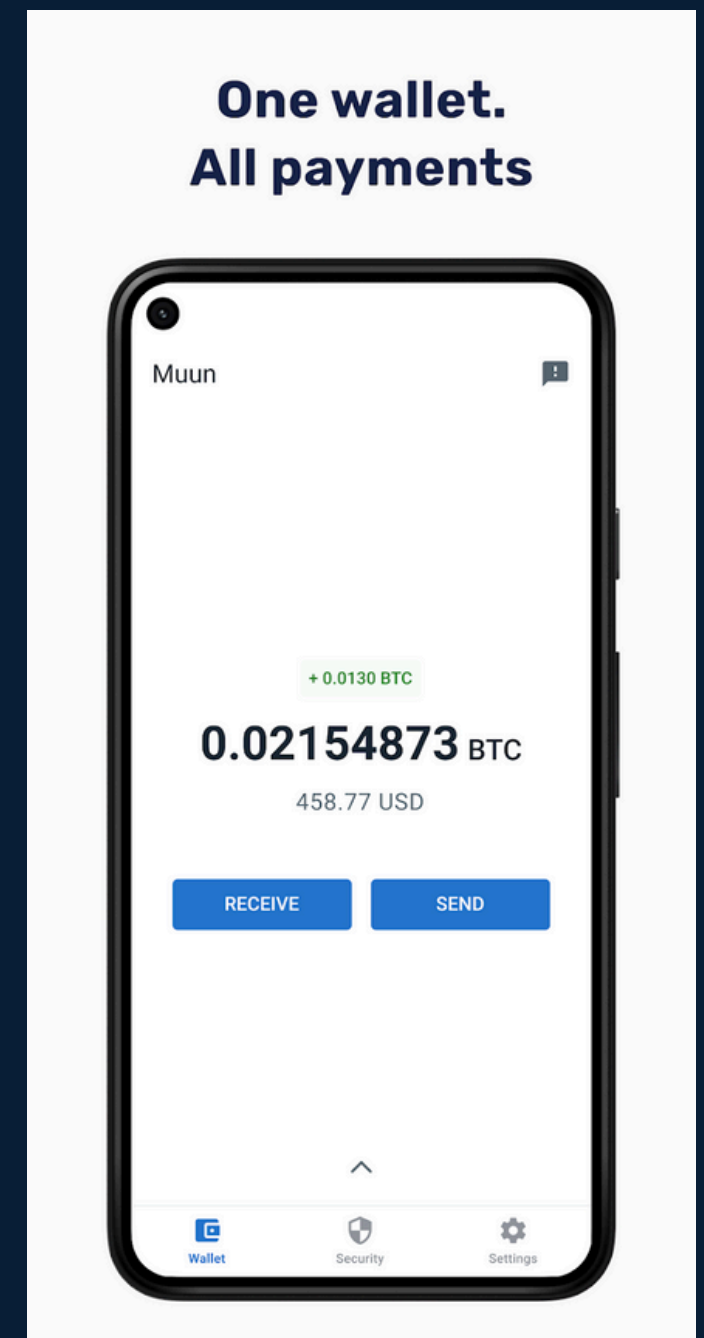
Liquidity

- Funds on other side of channel required to receive
- LSP = Lightning Service Provider
- Costs? Scalability?

A new way

Submarine Swaps

- Atomic Swaps with Lightning
- Pioneered by Muun
- No need to run a Lightning node



Connecting Bitcoin Layers



Lightning



Liquid



Rootstock



Arkade

Mainchain

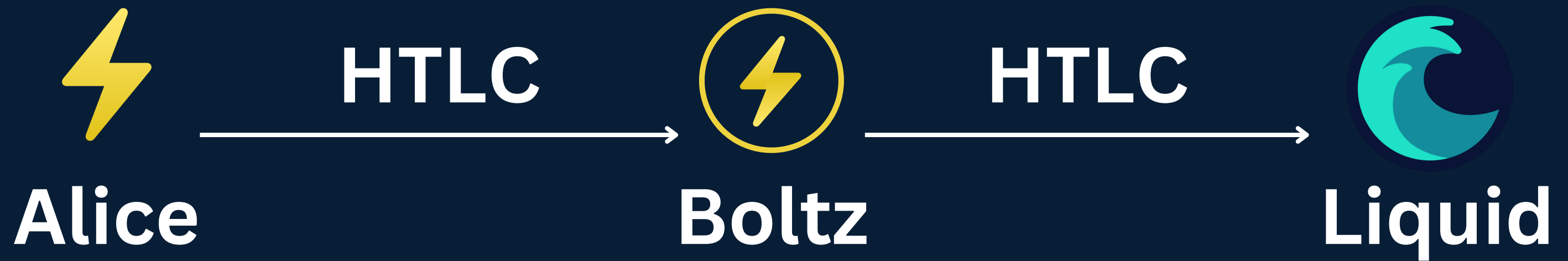
How?

HTLCs

What do we need?

- Hash lock
- Time lock

Swaps with Lightning



SegWit V0

```
OP_SIZE
OP_PUSHBYTES_1 20
OP_EQUAL
OP_IF
    OP_HASH160
    OP_PUSHBYTES_20 <preimage hash>
    OP_EQUALVERIFY
    OP_PUSHBYTES_33 <user public key>
OP_ELSE
    OP_DROP
    OP_PUSHBYTES_3 <timeout block height>
    OP_CLTV
    OP_DROP
    OP_PUSHBYTES_33 <Boltz public key>
OP_ENDIF
OP_CHECKSIG
```


SegWit V0

Hash lock



```
OP_SIZE
OP_PUSHTO_1 20
OP_EQUAL
OP_IF
  OP_HASH160
  OP_PUSHTO_20 <preimage hash>
  OP_EQUALVERIFY
  OP_PUSHTO_33 <user public key>
OP_ELSE
  OP_DROP
  OP_PUSHTO_3 <timeout block height>
  OP_CLTV
  OP_DROP
  OP_PUSHTO_33 <Boltz public key>
OP_ENDIF
OP_CHECKSIG
```

SegWit V0

Hash lock



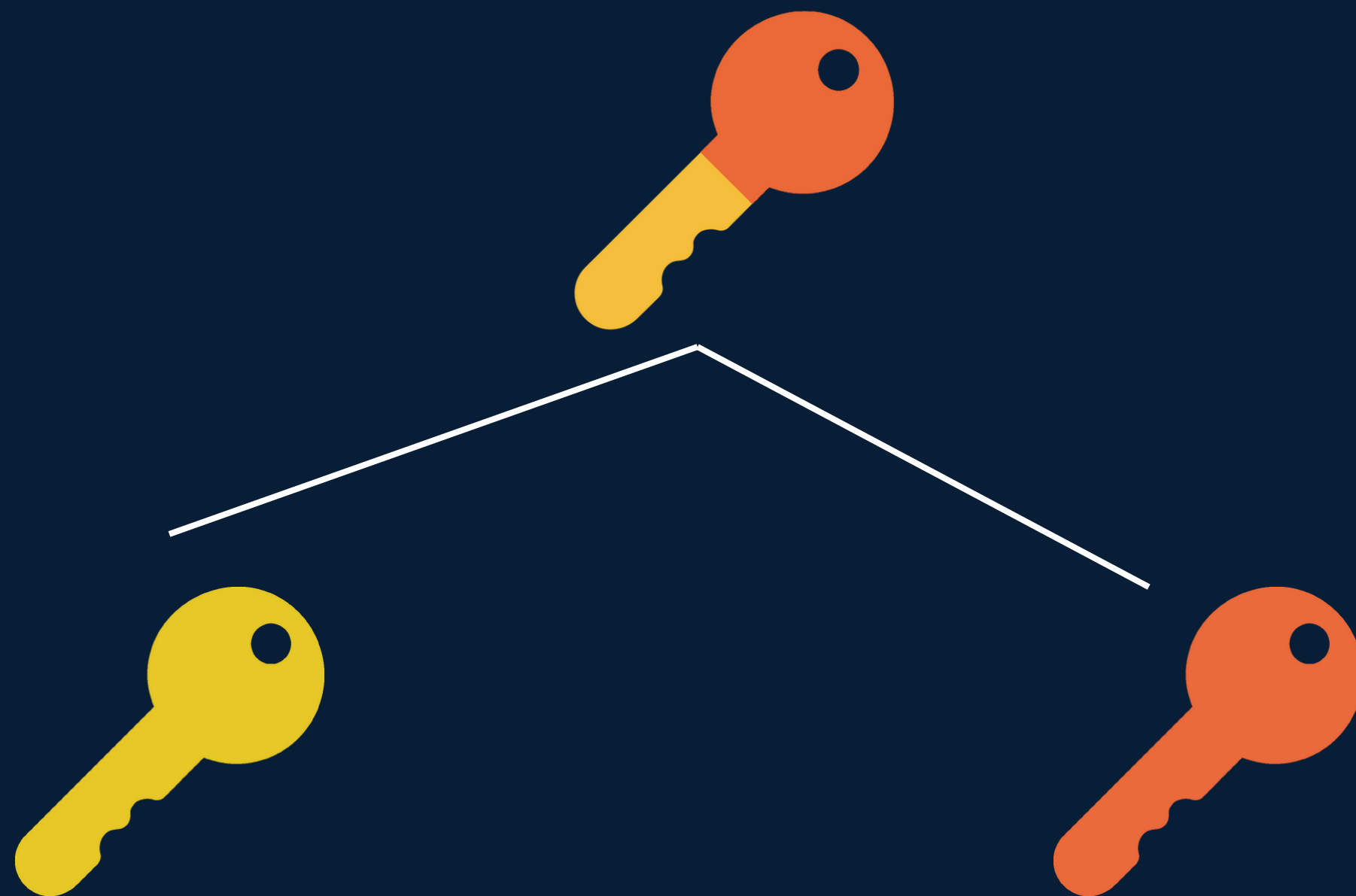
```
OP_SIZE
OP_PUSHBYTES_1 20
OP_EQUAL
OP_IF
  OP_HASH160
  OP_PUSHBYTES_20 <preimage hash>
  OP_EQUALVERIFY
  OP_PUSHBYTES_33 <user public key>
OP_ELSE
  OP_DROP
  OP_PUSHBYTES_3 <timeout block height>
  OP_CLTV
  OP_DROP
  OP_PUSHBYTES_33 <Boltz public key>
OP_ENDIF
OP_CHECKSIG
```

Time lock



Taproot

Musig2



Taptree



```
graph TD; Taptree --> HashLock[Hash lock]; Taptree --> TimeLock[Time lock];
```

Hash lock

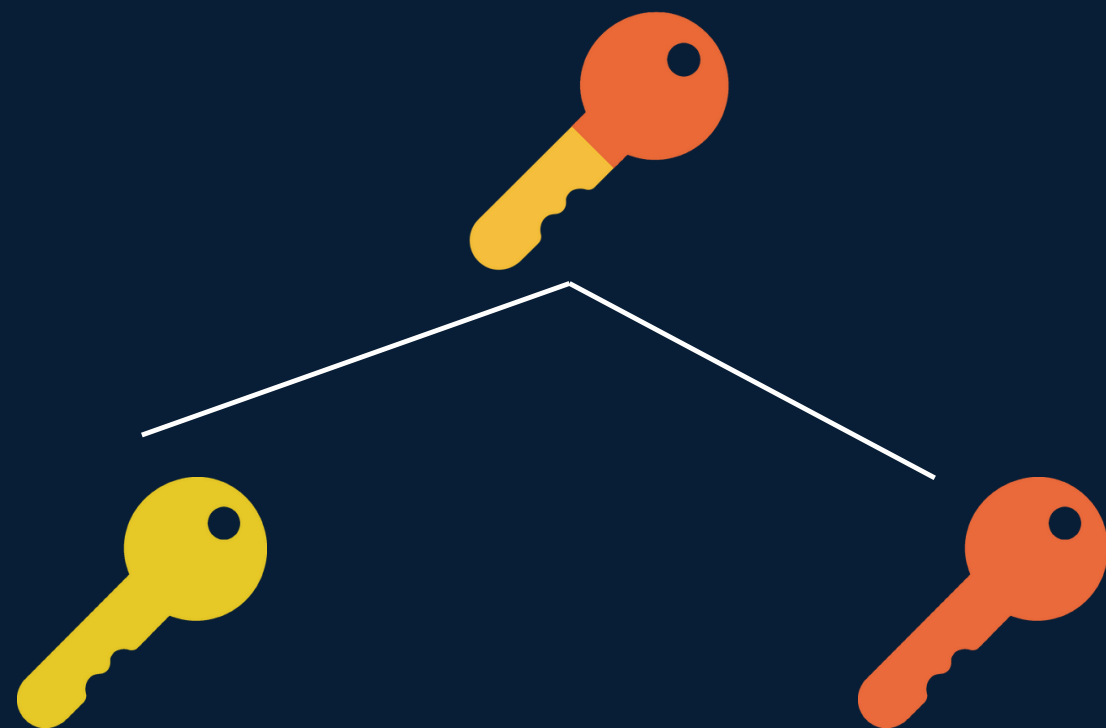
```
OP_SIZE  
OP_PUSHBYTES_1 20  
OP_EQUALVERIFY  
OP_HASH160  
OP_PUSHBYTES_20 <preimage hash>  
OP_EQUALVERIFY  
OP_PUSHBYTES_33 <user public key>  
OP_CHECKSIG
```

Time lock

```
OP_PUSHBYTES_33 <Boltz public key>  
OP_CHECKSIGVERIFY  
OP_PUSHBYTES_3 <timeout block height>  
OP_CLTV
```


Ways to spend

Key path



Script path

Hash lock

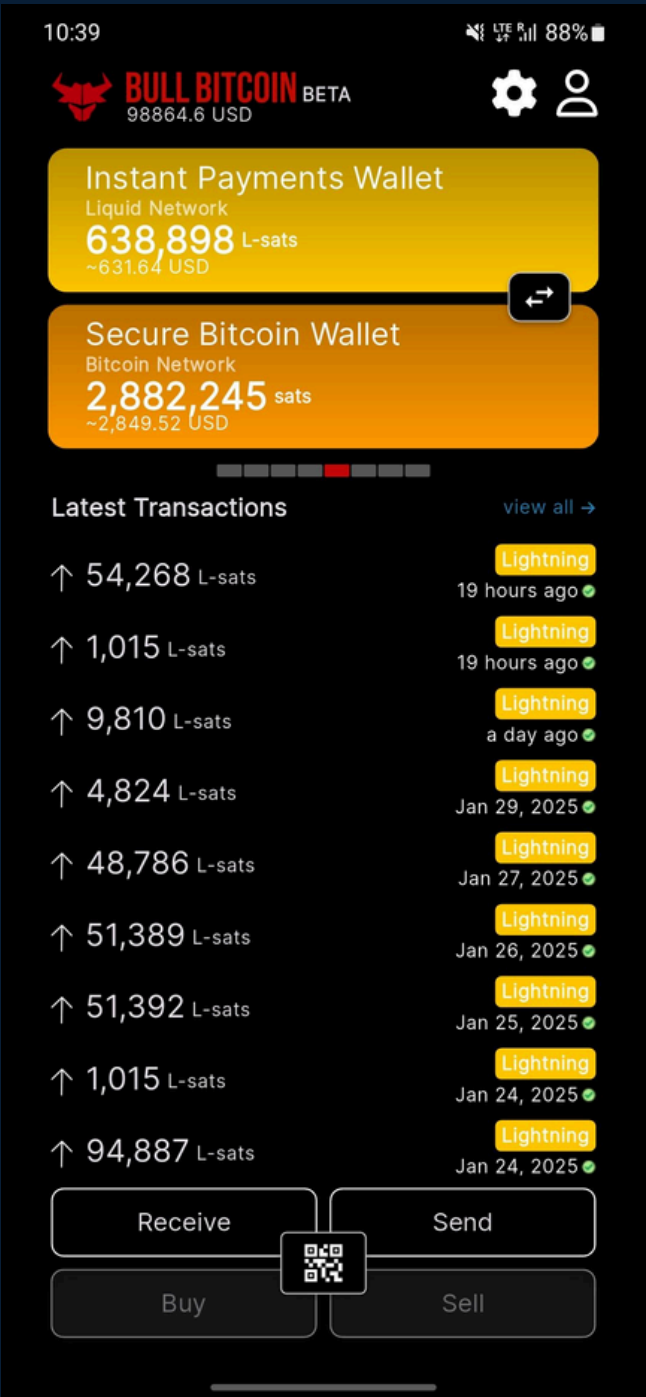
```
OP_SIZE  
OP_PUSHTO 1 20  
OP_EQUALVERIFY  
OP_HASH160  
OP_PUSHTO 20 <preimage hash>  
OP_EQUALVERIFY  
OP_PUSHTO 33 <user public key>  
OP_CHECKSIG
```

Time lock

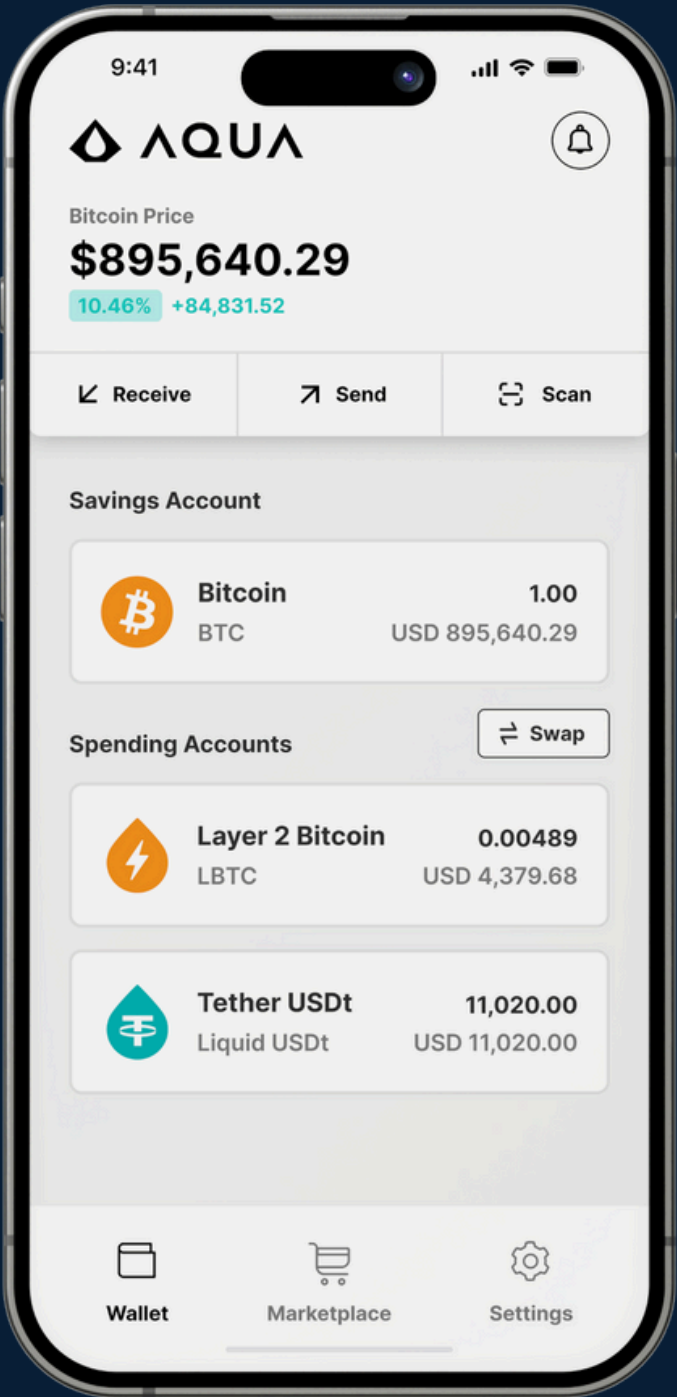
```
OP_PUSHTO 33 <Boltz public key>  
OP_CHECKSIGVERIFY  
OP_PUSHTO 3 <timeout block height>  
OP_CLTV
```

Wallets

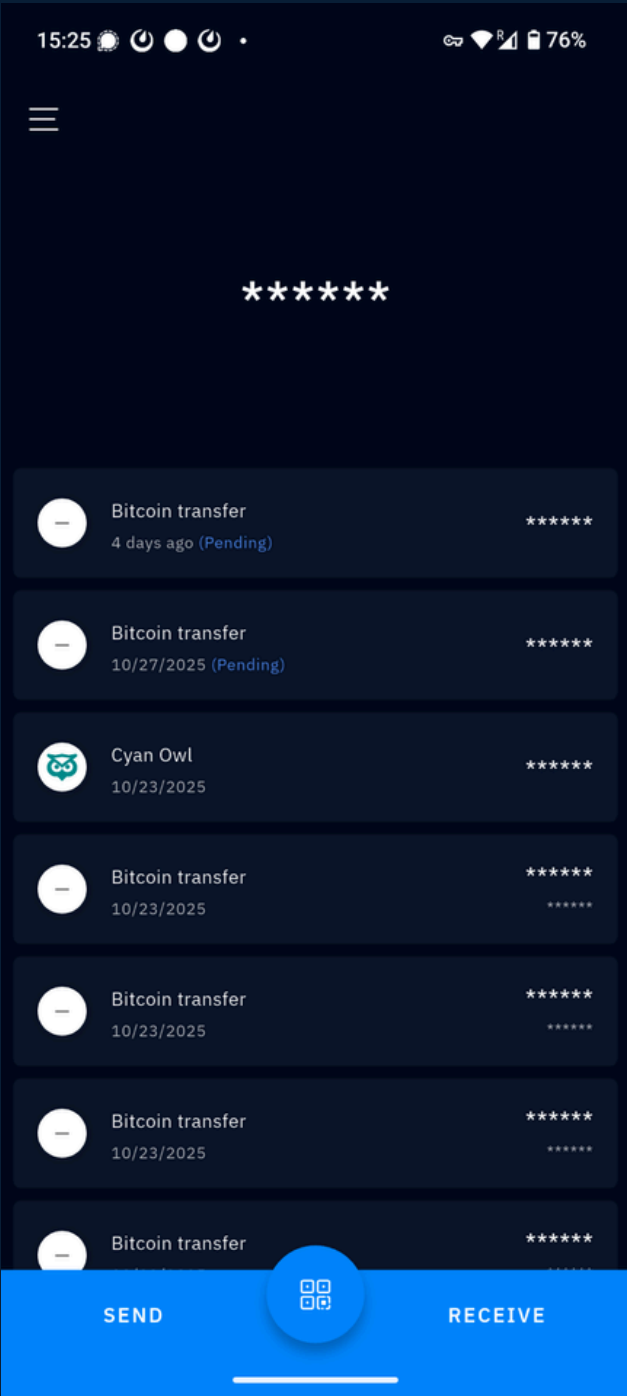
Bull Bitcoin



Aqua

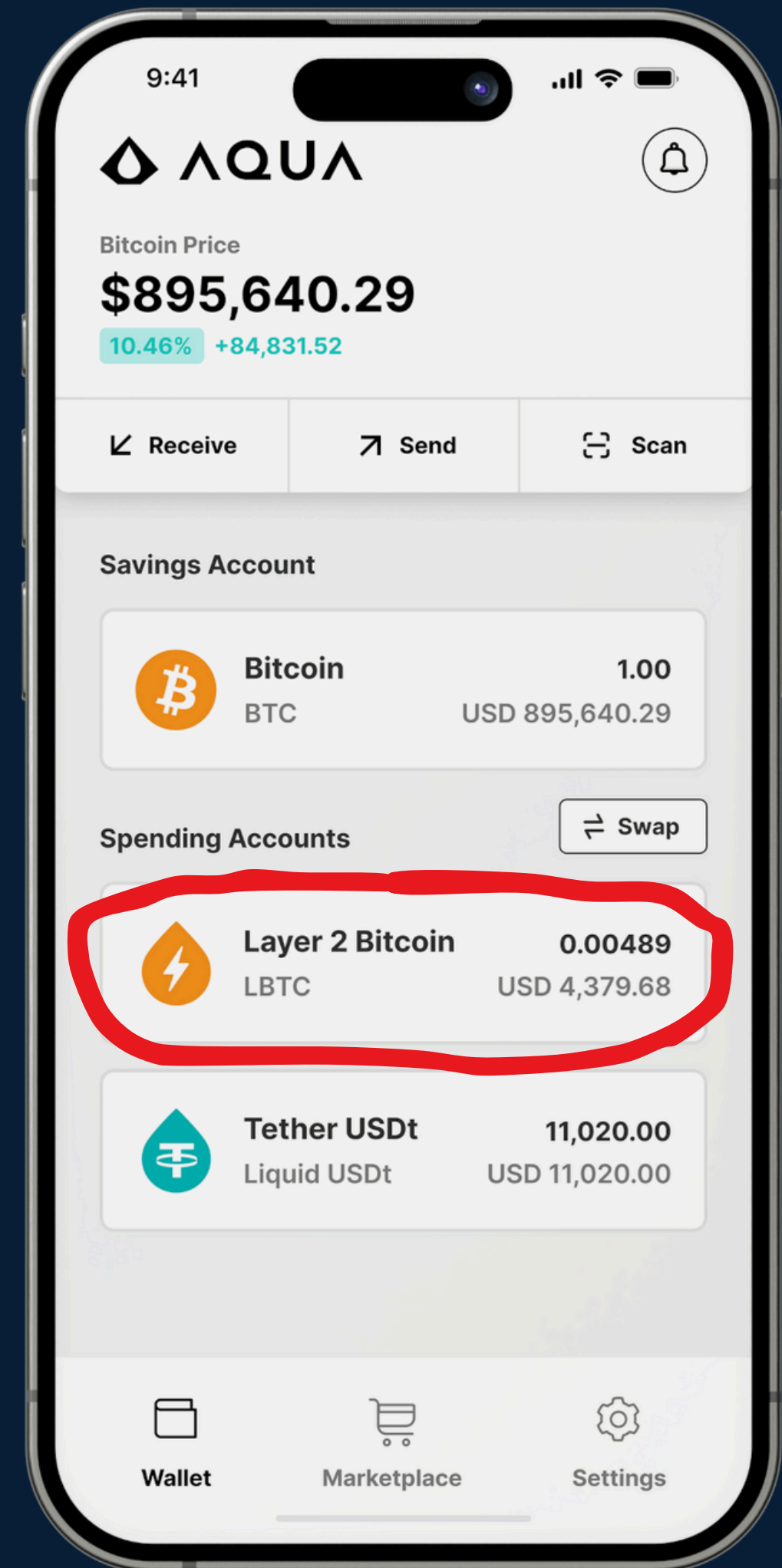


Misty Breez



How?

- Liquid Wallet
- Swap when needed



Why?

- Bitcoin
- Liquid
- Rootstock
- Ark
- Fedi
- Cashu

Lightning

Lightning Is the Common Language of the Bitcoin Economy



Roy Sheinfeld · Follow

Published in Breez Technology · 6 min read · Jun 12, 2024



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1



One of the best parts about running Breez is the diverse range of people I get to meet and work with. We have partners from [Jamaica](#), the [USA](#), [Switzerland](#), [Germany](#), [Canada](#), [Estonia](#), and who knows where else. We have users in [Finland](#), [Wales](#), [Namibia](#), [India](#), and almost everywhere else. The people behind Breez are split across three continents and come from a broad range of national and ethnic backgrounds.

Agreeing on a communication platform (Telegram? Slack? Zoom? Discord?) sometimes takes a bit of coordination. What never needs coordination, though, is the *language* we use to communicate. It's always automatically English. For many of us, English is our second (or third, or fourth) language, and parts of it are baffling, but it doesn't matter. Every initial contact is in

A new frontier

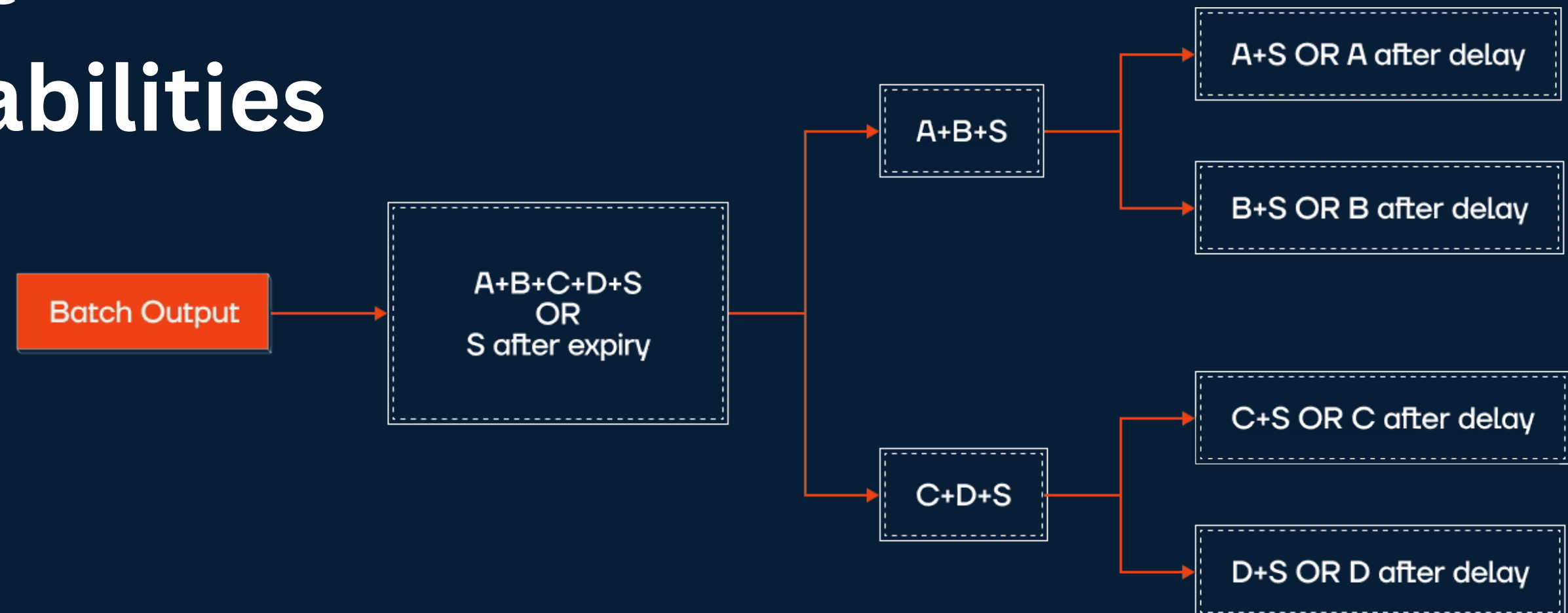
 **Arkade *Spark**

*Spark

- State chain
- Operators cosign
- Unilateral exit
- Lightning via Spark Service Provider



- UTXO batching engine
- Operator cosigns
- Unilateral exit
- Scripting capabilities





Lightning Swaps with HTLCs

Boltz

Connection status

<https://api.ark.boltz.exchange>✔ Connected

Swap history

	Arkade to Lightning ✔ - 391 SATS	Successful Nov 7, 2025, 02:41 AM
	Lightning to Arkade ✔ + 29,997 SATS	Successful Nov 7, 2025, 02:39 AM
	Lightning to Arkade ✖ + 2,499 SATS	Failed Nov 2, 2025, 02:30 PM
	Arkade to Lightning ✔ - 2,451 SATS	Successful Oct 31, 2025, 11:49 AM
	Lightning to Arkade ✔ + 2,499 SATS	Successful Oct 31, 2025, 11:47 AM